

Determining Solutions to InequalitiesWorksheet 7A.01.08 **ANSWER KEY**

Use this table of possible values for questions 1 and 2.

-10.5	0	$6\frac{3}{8}$	-4.999	-20.6
42	-5	12.75	$-1\frac{6}{17}$	-7.25

1. Which values from the table are solutions to $-25 < 4x - 5$?

$-25 < 4x - 5$ means $-20 < 4x$, so $x > -5$.

Solutions: -4.999, $-1\frac{6}{17}$, 0, $6\frac{3}{8}$, 12.75, 42

Six of the ten. -4.999 is a solution and -5 is not — the difference is one thousandth, and that is the point of putting them in the same table.

2. Which values from the table are solutions to $-25 > 4x - 5$?

$-25 > 4x - 5$ means $-20 > 4x$, so $x < -5$.

Solutions: -20.6, -10.5, -7.25

Three of the ten. Notice that -5 appears in neither list.

3. Rewrite both inequalities so that -5 is a solution to each. Explain your reasoning.

$-25 \leq 4x - 5$ and $-25 \geq 4x - 5$

$4(-5) - 5 = -25$ exactly, so -5 sits on the boundary. Changing $<$ to $\leq$ and $>$ to $\geq$ lets the boundary itself count as a solution.

4. Complete the sentence. An inequality has _____ solutions.

infinitely many

Every value on the shaded side of the number line works, and between any two of them there is another.

5. Complete the sentence. When you multiply or divide both sides of an inequality by a negative number, the inequality symbol _____.

reverses (flips to face the other way)

6. Tanner and Jonah were solving the inequality $-3 - y/7 < -5$. Jonah says the solution is $y < -14$. Tanner says the solution is $y > -14$. Do you agree with one of them, or with neither? Justify your answer mathematically.

Neither is correct.

$$-3 - y/7 < -5$$

$$-y/7 < -2 \quad (\text{add 3 to both sides})$$

$$y > 14 \quad (\text{multiply both sides by } -7 \text{ and reverse the symbol})$$

Check: $y = 21$ gives $-3 - 3 = -6 < -5$ ✓, and $y = 0$ gives $-3 < -5$ ✗

Both students kept the negative in the boundary. Multiplying -2 by -7 gives $+14$, not -14 . Jonah also failed to reverse the symbol; Tanner reversed it but carried the wrong sign.

For questions 7 through 10, solve the inequality, then graph the solution set on the number line provided.

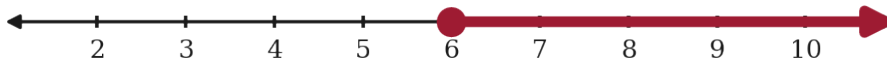
7. $5 + \frac{2}{3}b \geq 9$

$\frac{2}{3}b \geq 4$ (subtract 5 from both sides)

$b \geq 6$ (multiply both sides by $\frac{3}{2}$)

Check: $5 + \frac{2}{3}(6) = 5 + 4 = 9 \geq 9 \checkmark$

Solution. $b \geq 6$



Closed circle at 6 — the symbol is $\geq$, so 6 itself is a solution.

8. $4.6c - 9.5 < 4.3$

$4.6c < 13.8$ (add 9.5 to both sides)

$c < 3$ (divide both sides by 4.6)

Check: $c = 0$ gives $-9.5 < 4.3 \checkmark$, $c = 3$ gives $4.3 < 4.3 \times$

Solution. $c < 3$



Open circle at 3. Nothing was multiplied or divided by a negative, so the symbol does not move.

9. $-1 \leq -4 - y/6$

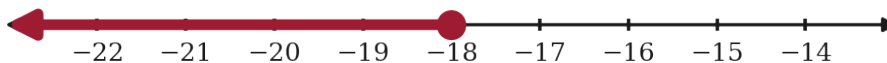
$3 \leq -y/6$ (add 4 to both sides)

$-18 \geq y$ (multiply both sides by -6 and reverse)

$y \leq -18$

Check: $y = -18$ gives -1 , and $-1 \leq -1 \checkmark$; $y = 0$ gives -4 , and $-1 \leq -4 \times$

Solution. $y \leq -18$



Writing the answer as $y \leq -18$ rather than $-18 \geq y$ makes the graph easier to read. Same statement either way.

10. $-7x - 12 > 16$

$-7x > 28$ (add 12 to both sides)

$x < -4$ (divide both sides by -7 and reverse)

Check: $x = -5$ gives $35 - 12 = 23 > 16 \checkmark$, $x = 0$ gives $-12 > 16 \times$

Solution. $x < -4$



Dividing by -7 reverses the symbol. Checking one value inside and one outside is what catches a missed flip.